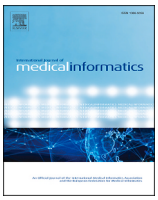
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Review article



De-identification of clinical data: A systematic review of free text, image and tabular data approaches

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HIGHLIGHTS

- Research in the field has seen a trend towards adopting new Large Language Models.
- De-identification works have focused on free text, with less emphasis on tabular data and even less on images.
- Challenges include de-identifying documents written in multiple languages.

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ABSTRACT

Background: The digitisation of healthcare has generated vast amounts of data in various formats, including free-text notes, tabular records and medical images. This data is critical for research and innovation, but often contains sensitive information that must be de-identified to ensure patient privacy and regulatory compliance. Natural Language Processing (NLP) enables automated de-identification of sensitive information to safely share medical datasets.

Objective: This study aims to systematically review the literature on NLP-based de-identification techniques applied to free-text medical reports, tabular data, and burned-in text within medical images over the past decade. It seeks to identify state-of-the-art methods, analyse how de-identification tasks are assessed, and find existing gaps for future research.

Methods: We systematically searched five important databases (PubMed, Web of Science, DBLP, ACM and IEEE) for articles published from January 2015 to December 2024 (10 years) about de-identification of medical data in free text, tabular data and burned-in pixels in images. We filtered the articles based on their titles and abstracts against inclusion and exclusion criteria, followed by a quality filter.

Results: From a set of 734 papers, 83 articles were deemed relevant. Most studies de-identify free text, with a few working with tabular data and a much scarcer number dealing with text embedded in the pixels of the images.

Conclusions: De-identification techniques have evolved, with increased use of Language Models and a decline in recurrence-based neural networks. Off-the-shelf tools often require customisation for optimal performance. Most studies de-identify English content, supported by the prevalence of English datasets. Key challenges include the phenomenon of code-mixing (i.e., more than one language used in the same sentence) and the scarcity of available datasets for reproducibility.

1. Introduction

The widespread adoption of technology in healthcare has increased the concern for medical data privacy. This data, encompassing free-text clinical notes, structured tabular records, and medical images, is invaluable for advancing research, improving patient outcomes and developing machine learning models to assist diagnoses [4,8,9]. However, its sensitive nature poses significant privacy concerns, necessitating techniques to ensure regulatory compliance [1–3].

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De-identification is essential to share medical data safely. Terminology is not unified in the literature. We follow the survey of Kovačević et al. [5] and adopt *de-identification* for detection, replacement or removal of Protected Health Information (PHI) regardless of the terms used in the reviewed papers. Amongst the various approaches to de-identification, Natural Language Processing (NLP) — which is increasingly widely used in the biomedical space [137,138] — is used to identify sensitive information in electronic health records (EHRs) while retaining the utility of the data for downstream tasks.

Previous systematic reviews of NLP techniques on medical data have focused solely on PHI found in free text reports [5–7,10]. However, sensitive text can be found in other types of documents. For example, medical data are typically stored in DICOM files [11]. They contain a greyscale image and associated pre-defined tabular metadata. Some fields can directly identify a patient, whereas others might help reveal their identity. Privacy issues are also found in images because text may be burned into their pixels.

This systematic review examines NLP-based de-identification methods for clinical notes, tabular data, and text embedded in images, summarising recent methods, datasets, and evaluation metrics. It updates and extends prior surveys that focused only on free text [5] or older surveys on broader types of de-identification that are now outdated [12], particularly in light of rapid NLP advances such as large language models.

2. Methods

2.1. Overview

We followed the systematic procedure of [13]: defined RQs, searched databases, applied selection criteria, and extracted data for analysis.

2.2. Research questions

1. Which de-identification tools and techniques are used per domain (text, image, tabular) and how have they evolved?
2. What shortcomings exist in the de-identification of language-specific medical data?
3. What are the current challenges for each domain (text, image, tabular)?

2.3. Searching

We searched five databases from medical and computing venues: ACM,[1] DBLP,[2] IEEE Digital Library,[3] ISI Web of Science[4] and PubMed.[5] ACM, DBLP and IEEE provide an index to computer science papers. PubMed is a database for biomedical studies, and Web of Science is a generalist index of publications.

Our query follows [5,14] and has three parts: one set of clauses related to *de-identification*, another for *clinical data* and the last for the *domain* of such data. The query is the same for all databases with just slight syntax adjustments specific to each platform (see Table 4). The general query is shown below: *("de-identification" OR "de-identifying") AND (clinical OR EHR OR EMR OR PHI OR medical) AND (NLP OR text OR "machine learning" OR tabular OR DICOM OR image OR OCR OR burned)*

The query was applied to the databases from January 2015 to December 2024 (10 years).

2.4. Inclusion and exclusion criteria

Our exclusion and inclusion criteria filtered relevant papers from the total number of studies returned by the search query. The following were applied to titles and abstracts:

Inclusion:

1. Study de-identifies burned-in pixels.
2. Study de-identifies tabular data.
3. Study de-identifies text.

Exclusion:

1. Surveys or positions.
2. Article is not about de-identification.
3. Data is not medical.
4. Article ≤ 3 pages.
5. Unavailable article.
6. Article is not written in English.
7. Data is in other formats (e.g., time series).
8. Study de-identifies face.

We excluded facial de-identification [15] (e.g., blurring images) to keep only text approaches. Although we excluded papers not written in English, we retained studies that de-identify text in other languages. **As quality filters**, we kept only articles published in venues indexed by CORE[6] or Scimago[7] rankings and only studies where automatic de-identification was the primary focus. Fig. 1 summarises our workflow.

3. Results

As shown in Fig. 1, our search query returned 734 unique works. After applying the filters, we ended up with a final set of 83 papers. They are listed in Table 5.

3.1. Bibliographic details

We report bibliographic details for the final set. Fig. 2(a) shows that most papers were published in the last five years, with peaks in 2020 and 2017. The dip in 2021 may have been impacted by the pandemic, but it has since shown a steady recovery. Fig. 2(b) counts author affiliations by country (papers often have authors from different countries, hence the total number is larger than the number of papers); the United States is most prevalent, which in part explains the common use of HIPAA (American Health Insurance Portability and Accountability Act)-style PHI definitions amongst the papers.

3.2. Findings: tabular data

In medical research and clinical practice, imaging data is usually in the form of DICOM image files with related non-image data provided as tabular data. The DICOM files themselves contain structured metadata as a series of key-value pairs. For the purposes of this paper, we are referring to both the DICOM headers and non-DICOM data as tabular data.

The DICOM supplement 142 [16] underpins several works [17–19] and lists tags with sensitive information (e.g., (0010,2150) "Country of Residence", (0010,1005) "Patient's Birth Name") to be deleted or replaced. As noted by Kundu et al. [18], it covers all 18 HIPAA categories. HIPAA classifies PHI into the categories shown in Table 1 (grouped into seven coarse classes) and is the most commonly used guideline [18–21]; the table also lists categories used by I2B2 and MIMIC, discussed in Section 3.3.

Structured DICOM metadata enables off-the-shelf de-identification tools. Aryanto et al. [17] evaluated ten free DICOM toolkits against 50 header tags from Supplement 142. Only DICOM Library[8] de-identified

[1] http://portal.acm.org.
[2] https://dblp.uni-trier.de/.
[3] http://ieeexplore.ieee.org.
[4] http://www.isiknowledge.com.
[5] https://www.ncbi.nlm.nih.gov/pubmed/.
[6] https://portal.core.edu.au/conf-ranks/.
[7] https://www.scimagojr.com/.
[8] https://www.dicomlibrary.com/.

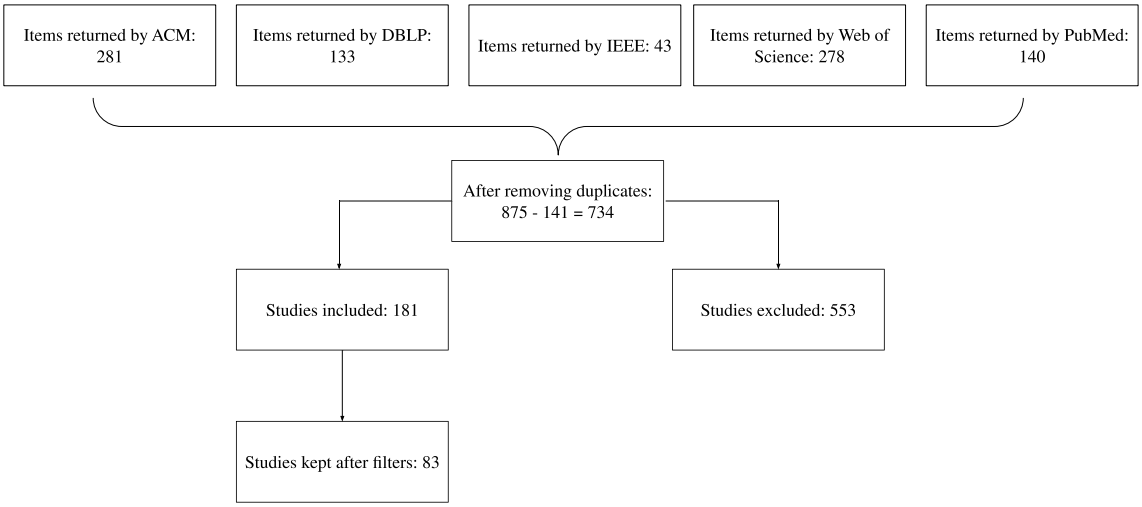
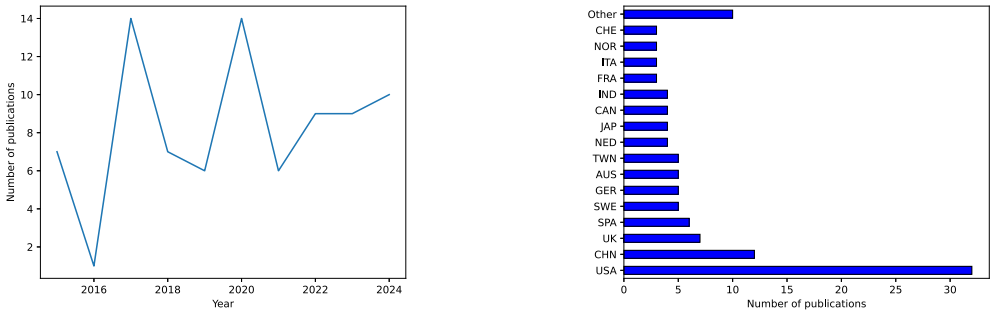


Fig. 1. Summary of the selection of studies.



(a) Publication count by year.

(b) Publication count by country affiliation.

Fig. 2. Bibliographic information.

all 50 by default; four others (GDCM,[9] PixelMed,[10] TudorDICOM,[11] and Yakami DICOM tool[12]) could after customization. Tests were limited to a single DICOM image.

Tabular data also appears in formats other than DICOM (e.g., relational databases), so de-identification must handle varied formats, field names and languages. Off-the-shelf tools may be language or format specific (e.g., Pantazos et al. [20] built a custom rule-based model for Danish EHR on Microsoft Axapta). Although DICOM defines standard tags, private tags can hold sensitive data, so automatic detection is required. For instance, Zhang and Jiang [21] combined rules with a gradient-boosted tree on 37 metadata features to identify sensitive tags.

An extreme approach from [23] replaces all tag values with synthetic ones: Generative Adversarial Networks (GANs) [24] can synthesise realistic tabular data without obvious leakage, but synthetic tables do not substitute for the analysis of original data. Evaluation typically uses precision, recall (sensitivity) [19,21,25], F-score [21,23] or a confusion matrix [20]. High recall is prioritised but can inflate false positives and reduce utility. Reported performance is generally high (typically >90 % for both precision and recall) [17,20,21].

3.3. Findings: free text

Due to the natural restrictions on sharing medical data, the few public datasets are widely adopted by several works. Datasets built for shared tasks are the most common. We summarise them in Table 2. They all have limited size, and therefore studies often need to collect private datasets with partner institutions. In these cases, however, the raw data cannot be shared.

The 2006 [26] and 2014 [27] I2B2 datasets and the 2016 CEGS NGRID [28] are the most used (see Table 5). I2B2 has eight classes (six HIPAA-aligned: patient, location, date, ID, phone, age); the latter two are larger and include non-HIPAA categories. The actual PHI words were replaced by artificial words. At the time of this writing, they are no longer available.[13] Outside English, the MEDDOCAN dataset contains synthetic Spanish documents. Built for the IberLEF2019 shared task [29], it is available in Huggingface.[14] It contains 1000 clinical cases with around 33,000 sentences. Another popular dataset is MIMIC-III [30], holding records of over 40,000 patients and is de-identified with categorised entity tags. Works have circumvented this limitation by replacing category tags with random values [31], or considering other types of words to be identified. For example, Bonomi et al. [34] used ICD9 codes for diseases and assessed “the effectiveness of several obfuscation methods in protecting these sensitive conditions”. Others, such as Ahmed et al. [32], Sarkar et al. [33] used ICD9 codes as a means to evaluate the utility

[9] http://gdcm.sourceforge.net/wiki/index.php/Main_Page.
[10] http://www.pixelmed.com/.
[11] http://santec.tudor.lu/project/optimage/dicom/start, however the link is not valid anymore.
[12] http://www.kuhp.kyoto-u.ac.jp/%7Ediag_rad/intro/tech/dicom_tools.html., also not valid anymore.

[13] https://portal.dbmi.hms.harvard.edu/projects/n2c2-nlp/.
[14] https://huggingface.co/datasets/GuiGel/meddocan.

Table 1
PHI types as defined by HIPAA, as well as I2B2 and MIMIC datasets. The mark "-" means that there are two or fewer instances in the dataset. In the MIMIC dataset, some PHI types are mapped to others due to ambiguity; the customised type is shown instead of "x." Table is taken from [22].

PHI categories	PHI types	Descriptions	HIPAA	I2B2	MIMIC
AGE	AGE	Ages ≥ 90	x	x	x
		Ages < 90	x		
CONTACT	PHONE	Telephone numbers	x	x	x
	FAX	Fax numbers	x	x	PHONE
	EMAIL	Electronic mail addresses	x	x	
	URL	Uniform resource locators	x	–	
	IP ADDRESS	Internet protocol addresses	x	–	
DATE	DATE	Dates (month and day parts)	x	x	x
		Year		x	x
		Holidays		x	x
		Days of the week		x	
ID	IDNUM	SSNs, license numbers	x	x	x
	MEDICAL RECORD	Medical record numbers	x	x	IDNUM
	DEVICE	Vehicle/device identifiers	x	x	IDNUM
	HEALTH PLAN	Health plan numbers	x	–	IDNUM
	BIOID	Biometric identifiers/photos	x	–	
LOCATION	STREET	Street address	x	x	x
	CITY	City	x	x	LOC.-OTHER
	ZIP	Zip code	x	x	x
	STATE	State		x	x
	COUNTRY	Country		x	x
	LOCATION-OTHER	Other identifiable locations		x	x
	ORGANIZATION	Employers	x	x	
	HOSPITAL	Hospital name		x	x
		Ward name			x
NAME	PATIENT	Names of patients/family	x	x	x
	DOCTOR	Provider name		x	x
	USERNAME	User IDs of providers		x	
PROFESSION	PROFESSION	Profession		x	

Table 2
Summary of datasets used in shared tasks, including their sizes and PHI categories.

Dataset	# Documents	PHI Categories	Language
2006 I2B2	889	8 categories	English
2014 I2B2	1304	7 broad, 28 specific.	English
2016 CEGS N-GRID	1000	7 broad, 28 specific.	English
MEDDOCAN	1000	21 categories	Spanish

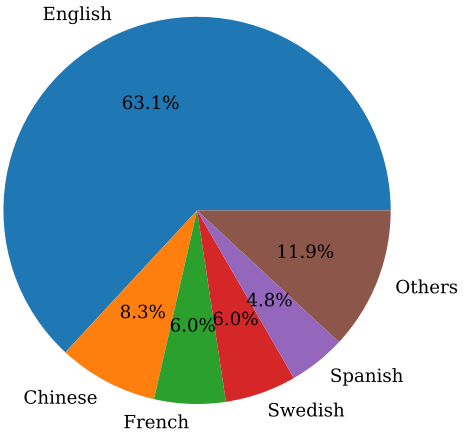


Fig. 3. Most common languages the documents were written.

of the text following obfuscation. Ideally, as Ahmed et al. [32] say, PHI "words do not affect the diagnosed disease as they are unimportant while making an ICD9 prediction".

The private medical datasets vary in size. Some reported hundreds of records [35–44], while others reported a few or tens of thousands [45–61]. A small number of works had access to larger datasets, making downsampling necessary [39,62–66].

Languages. Given that most publicly available datasets contain English texts, it is natural that most works de-identify English documents. In fact, most works deal with English documents, as Fig. 3 shows. A shared task that built a Spanish dataset facilitated work in this language [48, 67,68]. However, private datasets are the main source of data in other languages. In some cases, different papers report using the same private dataset, as for the Swedish language: the Stockholm Corpus [69] was utilised by a handful of works [44,70–73], but it contains real PHI and thus is not available for widespread use.

Another challenge arises when datasets contain more than one language. If the languages are linguistically close, it is possible to de-identify PHI in one using resources from the other, such as the case between Norwegian and Swedish, studied by Lamproudis et al. [72]. An alternative is to have an extensive list of rules [74].

The situation becomes more complicated when the same sentence is written with words in different languages, a phenomenon called code-mixing. This is particularly challenging for very different languages, such as English and Chinese. Wang et al. [60] illustrate this problem in Fig. 4: word segmentation systems may have trouble correctly segmenting text written in Latin and Chinese scripts. For these cases, multilingual language models (e.g., BERT) supporting large vocabularies have been used [60,65], or even Large Language Models (LLMs) [61]. However, performance metrics should be measured with care: Lee et al. [65] noticed that models identified PHI better in English sentences than in those

[張三]PATIENT is a [25]AGE-year-old male with history of …[張三]PATIENT was brought to [中國醫]HOSPITAL for help in [2095/8]DATE where he visited medical physician for his multiple somatic complaints such as muscle soreness and generalized weakness … education: studied at [中國醫]SCHOOL 獸醫三年級，因發病休學一年 …

Hospitalization:
Admission date R VS
[2105.12.15]DATE [王五]DOCTOR [趙六]DOCTOR

Fig. 4. Example taken from Wang et al. [60] depicting a code-mixed document.

mixed with Chinese characters. Wang et al. [60] observed that one PHI category where BERT performed the worst was for names, which is one of the most important categories to maximise recall. Moreover, Kotevski et al. [42] noticed that an off-the-shelf tool (Presidio[15]) struggled to identify Slavic, as well as Central and Southeast Asian names, compared to English names.

Still, there are tokens that may be PHI in one language, while having a different meaning in another. An example by Richter-Pechanski et al. [75] is "Die": it is an English verb, but it is also listed in the gazetteer "German and international first names". Thus, according to the context, the word may need to be removed, or kept for utility reasons.

Techniques: overall. Techniques fall into rule-based, machine learning, and hybrid approaches. From the machine learning models, neural networks with recurrence (e.g., Long Short-Term Memory — LSTM [76]) and Conditional Random Fields (CRF) [77] are the most widely adopted models. Recently, there has been a shift towards Transformers and LLMs. LLMs are in fact built on the Transformer architecture; when we use the term LLM, however, we refer specifically to large generative models with billions of parameters, such as ChatGPT [78] or LLaMA [79]. When we refer to Transformers, we refer to their earlier use in smaller models such as BERT and its derivatives [80,81].

Taking the data points in Fig. 5 and grouping them into four coarse-grained categories ((i) Off-the-Shelf tools, (ii) Traditional ML, (iii) Deep Learning, Transformers and LLMs as "Deep Learning", (iv) and the rest as Others), we fitted linear regression models with year of publication as independent variable and the number of papers as dependent variable, we find slopes to *quantify* trends. We find that the category with the largest increasing trend in adoption is Deep Learning (slope = 0.758), followed by Off-the-Shelf tools, with a modest slope of 0.248 (which are simpler to adopt as a baseline). Traditional ML and Others have declined with respective slopes of −0.085 and −0.11.

Fig. 5 shows that CRF usage peaked in 2020. CRFs are graph-based models that represent the conditional probability of a set of output labels given a set of input features. They then predict a class of words, provided that the words can be described in terms of features. Rule-based methods peaked in 2017 and have shown occasional spikes since. They remain common for non-English languages: over half of papers on Danish, Dutch, French, German, Japanese, or Korean used rule-based approaches (not always exclusively) [20,53,55,74,75,82,83].

The first neural-network de-identification study [22] used the LSTM architecture, highlighting that it does not require manual feature engineering and is robust to misspellings. That same year, Transformers were introduced [84], replacing recurrence with attention (built on earlier attention work [85]), and supplanted LSTMs.

Moreover, Transformers have increased in adoption at the same time that recurrence-based neural networks have declined. One particular Transformer architecture, BERT [80], has shown remarkable performance in numerous NLP tasks, and has been adopted by various works for PHI identification [49,51,55,61,72,86,87]. However, Transformers are no silver bullet: Chambon et al. [51] observed that they lose performance when tested in settings that differ from the ones they were trained on. Lastly, LLMs have begun to be used for de-identification,

[15] https://microsoft.github.io/presidio/.

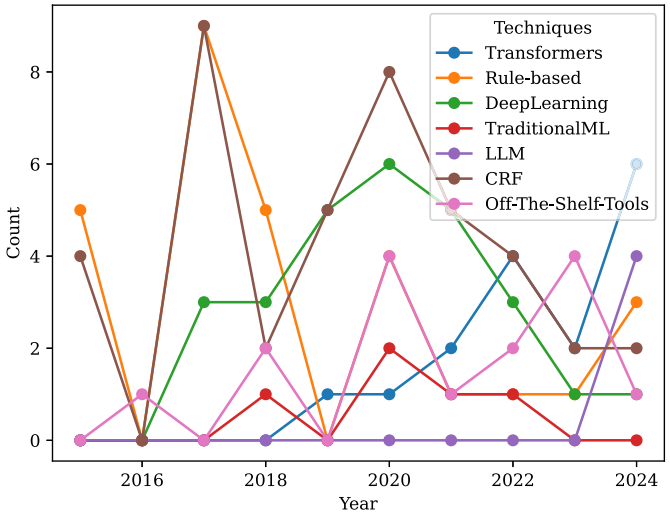


Fig. 5. Techniques used by papers in text de-identification through the years. DeepLearning refers to any neural network architecture other than Transformers (e.g., LSTM). We also make a distinction between LLM and Transformers: the former refers specifically to large generative models (e.g., ChatGPT), and the latter refers to smaller models (e.g., BERT).

Table 3
Off-the-shelf tools employed by at least two works.

Tool	Techniques	Used by
Philter [94]	Rule	[37,38,46,57]
NLM Scrubber [95]	Rule	[38,51,96]
MIT De-id [97]	Rule	[37,38,51,52,57,96,98]
MIST [99]	ML	[37,38,51,52,93,96,98]
NeuroNER [100]	ML	[38,51,57,93,96]
Emory HIDE [101]	ML	[51,96]
Presidio	Rule and ML	[42,102]

with four works appearing in 2024 [33,39,61,88], potentially starting a trend, given the rapid advancement of the field.

Works also propose hybrid solutions: Singh et al. [89] trained machine learning models, but also crafted a set of rules to deal with Maori names. Liu et al. [62] combined two CRF classifiers with a rule-based model, and their predictions were merged according to meta-rules prioritising the rule-based model. Alternatively, neural networks can be used to create features to be passed to CRFs [36,86,87,90–92].

Typically, papers report single-point F1/precision/recall scores without measures of variability or confidence intervals. In addition, *direct* comparison between metrics across different papers is problematic because of many factors, including different splits for common datasets, class imbalances across different databases and non standardised averaging (micro or macro).

Off-the-shelf tools. Off-the-shelf solutions can be treated as black-boxes, but internally they are typically either rule-based models and/or employ machine learning models. Table 3 describes the tools used for at least two papers. We also do not link Murugadoss et al. [93] for Philter, MIT De-id and Scrubber because the authors only report the results directly from previous papers without empirically assessing them.

ML-based models MIST, NeuroNER and Emory HIDE allow re-training [51] but can also be used out-of-the-box, although with limited reported performance [37,96]. Customisations may be required: [42] crafted a special set of regular expressions in Presidio to fit Australian formats for fields such as addresses and phone numbers.

The most used tools are MIST and MIT De-id. Heider and Meystre [38] evaluated six off-the-shelf tools and found that "no system uniformly outperformed the others at identifying PII across corpora and

categories." and also reported that "Deid and MIST have the largest variance between corpora". Another popular tool is Philter, a rule-based model that has reported the highest recall, but it requires "extensive revising of its pattern matching rules" according to El-Hayek et al. [37], who also concluded that off-the-shelf solutions struggle with particular types of PHI such as professions.

For de-identification to be effective, tools need to be adapted, which incurs costs. This cost may become more challenging in the near future should off-the-shelf tools incorporate large language models that require refined hardware.

Large language models. LLMs such as ChatGPT support different languages. Therefore, they can address one of the weaknesses of other models: rule-based models are inherently tied to the language they were designed to work with. Traditional machine learning techniques such as CRF often rely on language-dependent features. Even conventional neural networks need to be trained in different languages to recognise patterns in them. LLMs, in contrast, are already pretrained in different languages.

This fact was leveraged by Lee et al. [61], who compiled a clinical code-mixed de-identification dataset with documents in Chinese and English. In their data, the same sentence could contain words written in Latin or Chinese scripts. Compared to M-BERT [81], ChatGPT got better recall but lower precision in the zero-shot setting. This leads to the problem of prompt engineering to improve the performance of LLMs in identifying PHI. Given the variability that LLMs can have in replying the same prompt, Lee et al. [61], for instance, queried the LLM 3 times "to ensure stability of the responses".

The work of Yashwanth and Shettar [88] found that providing examples in the prompt helped LLMs to de-identify documents. The prompts also contained instructions about what should be de-identified. Particularly, the authors crafted a list with six rules for the LLM to follow, asking it to replace PHI with the token "[censored]". In the absence of examples or detailed instructions, the F1 score goes as low as.33, compared to.98 with a detailed prompt. GPT-3.5 and GPT-4.0 achieved the best results overall, whilst the open-source "LLaMA-7B did not perform as expected in de-identification". Furthermore, fine-tuning LLaMA "with 30 or 50 training examples does not make any difference in the performance", the authors say. This suggests that the size of LLMs plays an important role in their performance. Furthermore, LLMs modifications may harm utility. For instance, Langenbach et al. [39], who used a quantised LLaMA2 with 13 billion parameters, wrote that it was "prone to anonymize non-PHI information, did not achieve 100 % anonymization, and is therefore not recommended".

Alternatively, LLMs can create new synthetic documents from original notes, as done by Sarkar et al. [33]. However, this does not ensure that PHIs are not leaking, and the utility of the synthetic notes (evaluated in terms of downstream classification tasks) has to be further assessed. The same authors explain that "the scarcity of research work in downstream task-based evaluation is due to the fact that the usage of real clinical notes for a healthcare-related application is also scarce and has only begun to be investigated recently."

3.4. Findings: text in images

Fewer studies address text in images than in free text. Detecting burned-in text requires Optical Character Recognition (OCR) and often image preprocessing because strings are short and lack context that otherwise could help in discerning non-PHI from PHI (e.g., device serial number vs. patient SSN). Methods therefore extract all text from images and use auxiliary signals, such as matching OCR results to DICOM metadata, to decide which pieces of text are sensitive. Burned-in PHI is also rare; Vcelak et al. [103] found only 0.5 % of hospital files they had access to contained burned-in content.

One such early approach combining preprocessing techniques with the integration of auxiliary metadata was conducted by Monteiro et al. [104]. They extracted four DICOM attributes (name, ID, gender, accession) likely to be burned-in, preprocessed images with filters and used a CNN-OCR trained on the 73 K Character dataset [105]. To detect PHI, they adopted a method based on the Levenshtein distance between reconstructed words and words extracted from DICOM-sensitive fields.

Analogously, Vcelak et al. [103] processed nearly 7 million images from a Czech hospital and found that only 0.5 % contained PHI, so they manually curated and sampled the dataset. Preprocessing included greyscaling and binarisation; OCR was performed using the Tesseract tool [106]. A three-class classifier (no text/non-PHI text/PHI) achieved a recall of 94.85 %. Detection improved with higher image resolution and longer text strings.

The aforementioned works assume DICOM inputs, but data can be in other formats. For example, Jackson et al. [66] processed scanned PDFs (using Tesseract as an OCR engine) to extract addresses only, reporting up to 94 % precision and 96 % recall on a simulated 1000 test documents wrapped in "Lorem Ipsum" style.

Last, Demner-Fushman et al. [19] collected 8121 chest X-ray images from American hospitals and de-identified them with an off-the-shelf tool.[16] Humans checked for HIPAA identifiers and visible unique markers (implants, jaw/skull outlines, etc); 651 images contained at least one sensitive item.

4. Related surveys

A similar review to ours comes from Kovačević et al. [5], who analysed 69 papers only (2010–2023). Their review focused on de-identification for free English text and did not cover text in images or structured data, nor recent LLM-driven advances. Kushida et al. [12] surveyed 45 papers across different modalities but is now outdated, missing many neural-network–era advances. The earlier study of Meystre et al. [7] reviewed 18 studies (1995–2010) solely on free medical text. Despite its limited scope, it shows the beginning of the shift from rule-based models to machine learning.

Spasic and Nenadic [10] found that only 10 out of 110 studies had tens of thousands of documents, attributing small dataset sizes to costly manual annotation; crowdsourcing is generally unsuitable due to sensitive data. Leevy et al. [107] reported that ML models outperform rule-based ones, but warned that overfitting may inflate F-scores on public datasets, and found inconclusive benefits in combining deep learning with rule-based approaches.

Last, Hafsa and Kozeta [108] highlighted image-specific challenges such as face de-identification and synthetic image generation. NLP-related issues include de-identifying DICOM headers and detecting burned-in text. Their study, however, covered only eight papers.

Our study is more comprehensive, covering NLP de-identification across free text, tabular data, and text embedded in medical images, bridging gaps in prior surveys and highlighting key future research challenges.

5. Conclusions

This review summarises de-identification methods for clinical free text, tabular data, and text in images, answering our three RQs.

In response to RQ1, NLP has shifted towards Transformers and large language models; older recurrence-based networks are less used. Off-the-shelf tools often need customization. Image work is limited and typically combines OCR, preprocessing, and DICOM header checks. Tabular approaches use field masking or generative rewriting, but PHI can still hide in headers.

Regarding RQ2, most studies target English (reflected in available datasets). Rule-based methods remain common outside of English.

Last, we highlight the main challenges for RQ3: free text faces code-mixing and multilingual needs. Images suffer from few burned-in

[16] https://github.com/RSNA/Anonymizer.

examples and scarce public datasets. Tabular data face format variability (non-DICOM storage, relational schemas) that hinder interoperability.

CRediT authorship contribution statement

Pedro Faustini: Writing – original draft, Methodology, Data curation. **Annabelle McIver:** Writing – review and editing, Funding acquisition. **Ryan Sullivan:** Writing – review and editing, Funding acquisition. **Mark Dras:** Writing – review and editing, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Search strings

We show in Table 4 the specific syntax used in our search string for each database the works were searched. The conceptual search string is shown in Section 2.3.

Table 4
Search strings used in each database. Conceptually they are the same, but each database has peculiarities regarding syntax.

Databases	String
DBLP	(identif|deidentif)(clinical|EHR|EMR|PHI|medical)(NLP|text|machine learning|tabular|DICOM|image|OCR|burned)
Web of Science, IEEE, ACM	(“de-identification” OR “de-identifying” AND (clinical OR EHR OR EMR OR PHI OR medical) AND (NLP OR “text” OR “machine learning” OR “tabular” OR “DICOM” OR “image” OR “OCR” OR “burned”)
PubMed	(“de-identification” [Title/Abstract] OR “de-identifying” [Title/Abstract]) AND (clinical [Title/Abstract] OR EHR [Title/Abstract] OR EMR [Title/Abstract] OR PHI [Title/Abstract] OR medical [Title/Abstract]) AND (NLP OR “text” [Title/Abstract] OR “machine learning” [Title/Abstract] OR “tabular” [Title/Abstract] OR “DICOM” [Title/Abstract] OR “image” [Title/Abstract] OR “OCR” [Title/Abstract] OR “burned” [Title/Abstract])

Appendix B. Summary of works

We show in Table 5 the summary of all studies selected in our review. We report the domain of the study, the language of the data, metrics used for evaluation, datasets and which information was considered PHI.

Table 5
Summary of studies.

Study	Language	Domain	Technique	Metrics	Dataset	PHI
[109]	English	Text	Deep Learning	Precision, Recall, and F1	2006, 2014 I2B2 and 2016 CEGS N-GRID	28 categories
[110]	Korean, English	Text	Rules	Precision, Recall, F0.5 score	Private, 11,039 private clinical notes	21 categories
[35]	Italian	Text	Transformers, Deep Learning, CRF	Precision, Recall, and F1	Private, 115 covid notes	20 categories
[70]	Swedish	Text	CRF	Precision, Recall	Stockholm EPR PHI Corpus	8 classes: Age, Full date, Date part, First name, Last name, Health care unit, Location, and Phone number.
[45]	Chinese	Text	Rules and CRF	Precision, Recall, and F1	3 k documents from three Shanghai hospitals	Name, address, hospital
[46]	English	Text	Off-the-shelf tools	Precision, Recall.	18,000 manually annotated notes	HIPAA
[111]	English	Text	CRF and rules	Precision, Recall, and F1	2016 CEGS N-GRID	Name, Profession, Location, Contact, ID, Age, and Date, plus 30 subcategories
[47]	Chinese	Text	CRF and rules	Precision, Recall, and F1	14,719 discharge summaries from health centres in Ya’an	Name, date, age, location, institution, contacts, ids. Plus 15 subcategories
[36]	English	Text	Deep learning and CRF	Precision, Recall, and F1	2014 I2B2, and a private 500 clinical notes from Uni. Florida	10: date, name, age, id, phone, web, institute, street, city, zip
[104]	Unclear	Image	Image transformations, machine learning (linear models, neural networks, and random forest classifier)	Precision, Recall, and F1	Private	patient name, ID, gender and accession number.
[48]	Spanish	Text	Deep Learning and CRF	Precision, recall and F1	MEDDOCAN	29 categories

Table 5
Continued.

Study	Language	Domain	Technique	Metrics	Dataset	PHI
[49]	French	Text	Transformers	F1	3010 clinical documents from Lausanne University Hospital	9 categories (hospital unit, contact information, demographics, location, time, patient's ID, person's name, organization name, other personal information)
[112]	English	Text	Deep Learning and CRF	F1	I2B2 2014	Extended HIPAA categories
[50]	English	Text	Deep Learning	Accuracy, Precision and Recall	Private 5200 prostate cancer reports	Names
[113]	English	Text	Deep Learning and CRF	Precision, recall and F1	2014 I2B2 and 2016 CEGS N-GRID	HIPAA
[114]	Chinese	Text	Transformers and CRF	Precision, Recall, and F1	EHRs from Chongqing hospitals	organisation, location, dates, and personal names
[37]	English	Text	Off-the-shelf tools	Precision, Recall, and F1 and F2.	Private, 300 documents	7: Name, date, location, age, profession, contact, id
[38]	English	Text	Off-the-shelf tools	Recall, Precision, F2	Private 728 notes, 2006 and 2014 I2B2, 2016 CEGS N-GRID	HIPAA
[39]	English	Text	Rules and LLMs	Precision, Recall, and F1	Private. 100 chest CT reports	Dates, medical record numbers, ACCs, names
[51]	English	Text	Transformers, rules, off-the-shelf tools	Precision, Recall, and F1	999 chest X-ray and CT reports, 3001 X-rays and 2193 medical notes. Also 2006, 2014 I2B2	7: dates, providers, location, vendor, id, phone, patient name
[115]	French	Text	Deep Learning, Transformers and CRF	Precision, Recall, and F1	Rennes University Hospital: sampled 1000 hospital discharge summaries.	Patient, Doctor, Postal Address, Zip, City, Date, Email, and Phone Numbers
[62]	English	Text	Rules and CRF	Precision, Recall, and F1	2014 I2B2	7 main categories, 25 subcategories
[116]	English	Text	Rules and CRF	Precision, Recall, and F1	2014 I2B2	7 main categories, 25 subcategories
[93]	English	Text	Deep Learning, rules, CRF and off-the-shelf tools	Precision, Recall, and F1	2014 I2B2 and private 10 k physician notes	HIPAA plus others
[117]	English	Text	CRF	Precision, Recall, and F1	2014 I2B2	7 main categories, 25 subcategories
[25]	English	Tabular and text	Rules	Precision, Recall	synthetic database with 1000 fictional patients, 100 notes per patient	Patient id numbers, NHS numbers, names, birthdates, addresses, e-mail, phone, UK National Insurance numbers, family members'/ carers' names.
[91]	English	Text	Deep Learning and CRF	Precision, Recall, and F1	2014 I2B2	28 categories
[118]	English	Text	Rules and CRF	F1	2014 I2B2	7 main categories, 25 subcategories
[52]	English	Text	Deep Learning, rules, CRF and off-the-shelf tools	Precision, Recall, and F1	2016 N-GRID and 2014 I2B2	7: name, profession, location, age, date, contact, id
[92]	English	Text	Deep Learning and CRF	Precision, Recall, and F1	2006, 2014 I2B2 and MIMIC-III	HIPAA plus
[82]	Dutch	Text	Rules	Precision, Recall, and F1	200 nursing notes and 200 treatment plans from University Medical Center Utrecht	8: Name, location, institutions, date, ages, patient numbers, phone, email
[33]	English	Text	LLMs	AUC for privacy, F1 for utility	discharge summaries from the MIMIC-III dataset	Member / Non member in the training set
[75]	German	Text	Rules, off-the-shelf tools	Precision, Recall, and F2	Private cardiology documents	Date, location, person, phone, plz, salute, title
[119]	English	Text	CRF, Deep Learning and rules	Precision, Recall, and F1	2014 I2B2 and 2016 CEGS NGRID	grouped into 7 main categories
[90]	English	Text	Deep Learning and Transformers	F1	2014 I2B2 and CEGS NGRID 2016	grouped into 7 main categories
[32]	English	Text	Deep Learning, CRF	Precision, Recall, and F1.	2014 I2B2, MIMIC-II and MIMIC-III	HIPAA
[53]	French	Text	Rules	Recall, Precision	Private 11 k discharge summaries, sampled 20 for test	Dates, patient name, location, doctor name, nurse name, phone, address, companies
[22]	English	Text	Deep Learning	Recall, Precision, F1	2014 I2B2 and MIMIC-III	HIPAA grouped into 7 broad categories applied to both datasets
[18]	NaN	Tabular (DICOM Headers)	Rules	NaN	5 patients of head neck cancer	87 DICOM tags are selected from the DICOM supplement 142
[40]	English	Text	Deep Learning and Traditional ML	Recall, Precision, F1	Private, 600 discharge summaries	5 (person, address, dob, idn, phone)

Table 5
Continued.

Study	Language	Domain	Technique	Metrics	Dataset	PHI
[72]	Norwegian, Swedish and Danish	Text	Transformers	F1	6 private datasets	First Name, Last Name, phone, age, full date, date part, health care unit, location, organisation
[63]	Spanish	Text	Deep Learning and CRF	Recall, Precision, F1	Brain imaging radiology reports from the Medical Imaging Databank of the Valencian Region: 692 records	name, address, location, numbers, dates, institutions
[120]	Swedish	Text	Deep Learning and CRF	Recall, Precision, F1	Stockholm EPR PHI Pseudo Corpus	First Name, Last Name, phone, age, full date, date part, health care unit, location, organisation
[54]	Japanese	Text	Deep Learning, rules and CRF	Recall, Precision, F1	MedNLP-1 (500 sentences), dummy EHRs (32 patients), and pathology reports (1 K)	age, hospital, sex, time, and person
[41]	German	Text	Deep Learning and CRF	F2	113 cardiology reports	Person, Location, Date, Phone, Organisation, Title, Salute, ZIP code
[121]	English	Text	Rules and CRF	Recall, Precision, F1	2434 nursing notes from [97]	HIPAA
[122]	English	Text	CRF, rules and Traditional ML	F1	I2B2 and a private NZ dataset of 15 clinical documents	7: name, location, date, contact, profession, id, age
[55]	French	Text	Rule, CRF, Transformers	Recall, Precision, F1	Private dataset with 3682 documents	12 categories
[56]	French	Text	Transformers	Recall, Precision, F1	Private hospital data with 14,925 sentences.	HIPAA
[73]	Swedish	Text	Transformers	F1	Karolinska University Hospital and Stockholm EPR PHI Corpus.	9 categories: First Name, Last Name, Age, Phone Number, Location, Health Care Unit, Organization, Full Date, and Date Part
[34]	English	Text	Local DP graph-model	Maximum Attacker's Precision (MAP)	MIMIC III	45 ICD9 codes as sensitive conditions
[57]	English	Text	Off-the-shelf tools	Recall, Precision, F1	2503 radiology reports	HIPAA plus names of health-care workers, names of hospitals, and names of software/tools
[98]	English	Text	Deep Learning, CRF, Traditional ML and off-the-shelf tools	Recall, Precision, F1	2014 I2B2	7 categories
[123]	English	Text	Deep Learning and CRF	Precision, Recall, and F1	2010, 2012 and 2014 I2B2	
[42]	English	Text	Off-the-shelf tools	Recall, Precision, F1	Private, 300 documents	13 categories
[124]	English	Text	Off-the-shelf tools	Recall, Precision, F1	2503 radiology reports	21 categories
[43]	English	Text	Rules	Recall	473 Department of Veterans Affairs documents	Name, Location, Phone, Fax, SSN, Medical record number
[125]	English	Text	Transformers	F1	1334 private Nursing notes, 2014 I2B2	HIPAA, grouped into 10 categories: Date, Hospital, Location, Patient or Care Partner Name, Staff Name, Id, Phone, Age, Organization and Email
[17]	Unclear	Tabular (DICOM Headers)	Off-the-shelf tools	Success rate (not explained)	A dummy DICOM image	50 columns
[64]	Dutch	Text	Deep Learning, Transformers and CRF	Precision, Recall, and F1	Private EHRs of 39 healthcare Dutch organizations. Also, NUT corpus [126] with 1260 records.	Name, Date, Initials, Address, Care Institute, Organisation, Internal Location, Phone, Age, Email, Hospital, Profession, URL, ID (14 total).
[23]	English	Tabular	GANs	F1, AUROC	Epileptic Seizure (11,500 rows) [127], Thyroid (9100) [128], Diabetes (89 k) [129] MIMIC III (42,900)	
[67]	Spanish	Text	Off-the-shelf tools	Precision, Recall, and F1	MEDDOCAN	21, see its Table 1
[31]	English	Text	Off-the-shelf tools and CRF	Recall, F1, IL	2006, 2014 I2B2, MIMIC	
[103]	Czech, English	Image	Image transformations, OCR, classification	Recall	15,334 images	Address, age, birth, name, sex, accession number, description, academic degree
[58]	English	Text	CRF	Precision, Recall, and F1	1500 advanced (NSCLC) pathology reports from the University of Missouri Health Care	Name, date, hospital, location, phone, initials, id number
[130]	English	Text	Rules and CRF	Precision, Recall, and F1	2016 CEGS N-GRID	25 types

Table 5
Continued.

Study	Language	Domain	Technique	Metrics	Dataset	PHI
[131]	English	Text	CRF	Precision, Recall, and F1	Outpatient notes from the University of Texas Health Science Center, 2006 and 2014 I2B2, 2016 CEGS N-GRID	eight PHI categories, or Date, Doctor, Phone, Patient, Location, ID, Hospital, and Age.
[132]	English, Chinese	Text	Deep Learning	Precision, Recall, and F1	2006, 2014 I2B2, private Chinese EMRs with 9700 records	ID, date, hospital, doctor, patient, age
[133]	English	Text	Rules, off-the-shelf tool	Precision, Recall, and F1	2010 I2B2	HIPAA
[134]	English	Text	Deep Learning	Precision, Recall, and F1 between n-gram overlap original and synthetic notes (the lower the better)	5.5 million de-identified ED visit records provided by the New York City Department of Health and Mental Hygiene (NYC DOHMH).	Names
[59]	English	Text	Rules and Transformers	Precision, Recall, and F1	2100 pathology reports	Name, profession, Location, Age, Date, Contact, ID, Other
[19]	English	Text, Image, tabular	Off-the-shelf tools	Precision	8121 images and 3996 chest reports	HIPAA plus face characteristics
[20]	Danish	Tabular and text	Rules	Confusion matrix	Sample of 369 patient records from 79 Danish clinics and hospitals	civil reg. Number code, last name, first name, street, zip, hospital, date, country, phone, age, city (see Table 1)
[44]	Swedish	Text	CRF	PHI prevalence, Precision, Recall, and F1	100 health records from five clinics (Neurology, Orthopedics, Infection, Dental Surgery, and Nutrition)	Health Care Unit, First Name, Last Name, Date Part, Full Date, Location, Phone, Age
[60]	mixture of English and Chinese	Text	Transformers	Precision, Recall, and F1	1700 EHRs from NTUHIntegrated Medical Database	6 main categories and 25 subcategories
[65]	mixture of English and Chinese	Text	Rule, CRF, Transformers, Deep Learning	F1	4737 unstructured discharge summaries from the National Taiwan University Hospital, selected 800	HIPAA's 18 categories + 3
[68]	Spanish	Text	Transformers, CRF	Precision, Recall, and F1	MEDDOCAN and NUBes (7 k medical reports, private version with sensitive information)	NUBes: 11 categories, MEDDOCAN: 21
[21]	English	Structured data	Rules and Traditional ML	AUROC, AUPRC, accuracy, prediction, recall, and F1 score	8 EHR datasets. Unclear size (at least 5 k samples). Total 889 features	HIPAA
[135]	English	Text	Rules and CRF	Precision, Recall, and F1	I2B2 (2010/14/16)	20 categories
[61]	English, Chinese	Text	Transformers, rules and LLMs	micro/macro-F1 measure, precision and recall	1700 discharge summaries	See Table 1: Six main categories: Date. Age.Name, Location, Profession, ID. They also have 20 subcategories
[136]	English	Text	Token replacement via word embeddings	Recall, Pearson, Acc	Private	Not reported
[88]	English	Text	LLMs	entity-wise accuracy and F1 scores	2014 I2B2	HIPAA's 18 categories (But dataset has combined them into 6)
[66]	English	Text and images (scanned documents, not dicom)	CogStack (rule based on metadata and tesseract)	Precision, Recall	King's College Hospital private data, 1 k generated data for evaluation	Addresses

References

[1] T. Hwang, A. Kesselheim, K. Vokinger, Lifecycle regulation of artificial intelligence– and machine learning–based software devices in medicine, JAMA. 322 (12) (2019) 2285–2286, https://doi.org/10.1001/jama.2019.16842

[2] T. Minssen, S. Gerke, M. Aboy, N. Price, G. Cohen, Regulatory responses to medical machine learning, J. Law Biosci. 7 (lsaa002) (4 2020), https://doi.org/10.1093/jlb/lsaa002

[3] E. Petersen, Y. Potdevin, E. Mohammadi, S. Zidowitz, S. Breyer, D. Nowotka, S. Henn, L. Pechmann, M. Leucker, P. Rostalski, C. Herzog, Responsible and regulatory conform machine learning for medicine: a survey of challenges and solutions, IEEE Access 10 (2022) 58375–58418.

[4] M. Ahsan, S. Luna, Z. Siddique, Machine-learning-based disease diagnosis: a comprehensive review, Healthcare 10 (2022), https://www.mdpi.com/2227-9032/10/3/541.

[5] A. Kovačević, B. Bašaragin, N. Milošević, G. Nenadić, De-identification of clinical free text using natural language processing: a systematic review of current approaches, Artif. Intell. Med. 151 (2024) 102845, https://doi.org/10.1016/j.artmed.2024.102845, https://www.sciencedirect.com/science/article/pii/S0933365724000873.

[6] B. Negash, A. Katz, C.J. Neilson, M. Moni, M. Nesca, A. Singer, J.E. Enns, De-identification of free text data containing personal health information: a scoping review of reviews, Int. J. Popul. Data Sci. 8 (2023) 2153.

[7] S.M. Meystre, F.J. Friedlin, B.R. South, S. Shen, M.H. Samore, Automatic de-identification of textual documents in the electronic health record: a review of recent research, BMC Med. Res. Methodol. 10 (2010) 70, https://doi.org/10.1186/1471-2288-10-70.

[8] K. Kourou, T. Exarchos, K. Exarchos, M. Karamouzis, D. Fotiadis, Machine learning applications in cancer prognosis and prediction, Comput. Struct. Biotechnol. J. 13 (2015) 8–17, https://www.sciencedirect.com/science/article/pii/S2001037014000464.

[9] W. Yue, Z. Wang, H. Chen, A. Payne, X. Liu, Machine learning with applications in breast cancer diagnosis and prognosis, Designs 2 (2018), https://www.mdpi.com/2411-9660/2/2/13.

[10] I. Spasic, G. Nenadic, Clinical text data in machine learning: systematic review, JMIR Med. Inform. 8 (2020), https://doi.org/10.2196/17984, https://www.sciencedirect.com/science/article/pii/S2291969420001131.

[11] C. Parisot, The Dicom standard, Int. J. Card. Imaging 11 (1995) 171–177, https://doi.org/10.1007/BF01143137

[12] C.A. Kushida, D.A. Nichols, R. Jadrnicek, R. Miller, J.K. Walsh, K. Griffin, Strategies for de-identification and anonymization of electronic health record data for use in multicenter research studies, Med. Care. 50 (2012) S82–S101, https://www.scopus.com/inward/record.uri?eid=2-s2.0-84862293738&doi=10.1097.

[13] B. Kitchenham, Procedures for Performing Systematic Reviews, Keele University. Technical Report TR/SE-0401, Department of Computer Science, Keele University, UK, 2004.

[14] John Wiley & Sons, Ltd, Front matter, 2008, pp. i–xxi. https://onlinelibrary.wiley.com/doi/abs/10.1002/9780470712184.fmatter.

[15] Y. Wen, B. Liu, L. Song, J. Cao, R. Xie, Overview of Face De-Identification Techniques, Springer, Nature Switzerland, Cham, 2024, pp. 23–57. https://doi.org/10.1007/978-3-031-58222-6_3

[16] DICOM, Supplement 142: clinical trial de-identification profiles, supplement 142, DICOM standards committee, 2011. Available at https://www.dicomstandard.org/News/ftsup/docs/sups/sup142.pdf.

[17] K.Y.E. Aryanto, M. Oudkerk, P.M.A. van Ooijen, Free DICOM de-identification tools in clinical research: functioning and safety of patient privacy, Eur. Radiol. 25 (2015) 3685–3695.

[18] S. Kundu, S. Chakraborty, S. Chatterjee, S. Das, R.B. Achari, J. Mukhopadhyay, P.P. Das, I. Mallick, M. Arunsingh, T. Bhattacharyyaa, S. Ray, De-identification of radiomics data retaining longitudinal temporal information, J. Med. Syst. 44 (2020) https://doi.org/10.1007/s10916-020-01563-0

[19] D. Demner-Fushman, M.D. Kohli, M.B. Rosenman, S.E. Shooshan, L. Rodriguez, S. Antani, G.R. Thoma, C.J. McDonald, Preparing a collection of radiology examinations for distribution and retrieval, J. Am. Med. Inform. Assoc. 23 (2016) 304–310, https://doi.org/10.1093/jamia/ocv080

[20] K. Pantazos, S. Lauesen, S. Lippert, Preserving medical correctness, readability and consistency in de-identified health records, Health Inform. J. 23 (2017) 291–303, https://doi.org/10.1177/1460458216647760

[21] K. Zhang, X. Jiang, Sensitive data detection with high-throughput machine learning models in electrical health records, in: AMIA Annu. Symp. Proc. 2023, 2023, pp. 814–823.

[22] F. Dernoncourt, J.Y. Lee, O. Uzuner, P. Szolovits, De-identification of patient notes with recurrent neural networks, J. Am. Med. Inform. Assoc. 24 (2017) 596–606, https://doi.org/10.1093/jamia/ocw156

[23] A.H.Z. Nik, M.A. Riegler, P. Halvorsen, A.M. Storas, Generation of synthetic tabular healthcare data using generative adversarial networks, in: D. Dang-Nguyen, C. Gurrin, M. Larson, A. Smeaton, S. Rudinac, M. Dao, C. Trattner, P. Chen (Eds.), MULTIMEDIA MODELING, MMM 2023, PT I, Volume 13833 of Lecture Notes in Computer Science, SPRINGER INTERNATIONAL PUBLISHING AG, GEWERBESTRASSE 11, CHAM, CH-6330, Bergen, NORWAY, SWITZERLAND, 2023, 09-12 Jan 2023, pp. 434–446, https://doi.org/10.1007/978-3-031-27077-2_34, 29th International Conference on MultiMedia Modeling (MMM).

[24] I.J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, in: Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, K. Weinberger (Eds.), Generative adversarial nets, Advances in Neural Information Processing Systems, volume 27, Curran Associates, Inc. 2014, https://proceedings.neurips.cc/paper_files/paper/2014/file/f033ed80deb0234979a61f95710dbe25-Paper.pdf.

[25] R.N. Cardinal, Clinical records anonymisation and text extraction (CRATE): an open-source software system, BMC Med. Inform. Decis. Mak. 17 (2017) 50.

[26] O. Uzuner, Y. Luo, P. Szolovits, Evaluating the state-of-the-art in automatic de-identification, J. Am. Med. Inform. Assoc. 14 (2007) 550–563, https://doi.org/10.1197/jamia.M2444, https://academic.oup.com/jamia/article-pdf/14/5/550/2136261/14-5-550.pdf.

[27] A. Stubbs, Ö. Uzuner, Annotating longitudinal clinical narratives for de-identification: the 2014 i2b2/Uthealth corpus, J. Biomed. Inform. 58 (2015) S20–S29, https://doi.org/10.1016/j.jbi.2015.07.020, https://www.sciencedirect.com/science/article/pii/S1532046415001823, supplement: Proceedings of the 2014 i2b2/UTHealth Shared-Tasks and Workshop on Challenges in Natural Language Processing for Clinical Data.

[28] A. Stubbs, M. Filannino, Ö. Uzuner, De-identification of psychiatric intake records: overview of 2016 Cegs n-grid shared tasks track 1, J. Biomed. Inform. 75 (2017) S4–S18, https://doi.org/10.1016/j.jbi.2017.06.011, https://www.sciencedirect.com/science/article/pii/S153204641730134X, supplement: A Natural Language Processing Challenge for Clinical Records: Research Domains Criteria (RDoC) for Psychiatry.

[29] M. Marimon, A. Gonzalez-Agirre, A. Intxaurrondo, H. Rodriguez, J.L. Martin, M. Villegas, M. Krallinger, Automatic de-identification of medical texts in Spanish: the MEDDOCAN track, corpus, guidelines, methods and evaluation of results, in: M.Á.G. Cumbreras, J. Gonzalo, E.M. Cámara, R. Martínez-Unanue, P. Rosso, J. Carrillo-de-Albornoz, S. Montalvo, L. Chiruzzo, S. Collovini, Y. Gutiérrez, S.M. Jiménez-Zafra, M. Krallinger, M. Montes-Y-Gómez, R. Ortega-Bueno, A. Rosá (Eds.), Proceedings of the Iberian Languages Evaluation Forum Co-Located with 35th Conference of the Spanish Society for Natural Language Processing, IberLEF@SEPLN 2019, volume 2421 of CEUR Workshop Proceedings, Bilbao, Spain, CEUR-WS.org, 24 September 2019, 2019, pp. 618–638, https://ceur-ws.org/Vol-2421/MEDDOCAN_overview.pdf.

[30] A.E.W. Johnson, T.J. Pollard, L. Shen, L.-W.H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L.A. Celi, R.G. Mark, MIMIC-III, a freely accessible critical care database, Sci. Data 3 (2016) 160035.

[31] B. Ribeiro, V. Rolla, R. Santos, INCOGNITUS: a toolbox for automated clinical notes anonymization, in: D. Croce, L. Soldaini (Eds.), Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations, Dubrovnik, Croatia, Association for Computational Linguistics, 2023, pp. 187–194, https://doi.org/10.18653/v1/2023.eacl-demo.22

[32] T. Ahmed, M.M.A. Aziz, N. Mohammed, De-identification of electronic health record using neural network, Sci. Rep. 10 (2020) https://doi.org/10.1038/s41598-020-75544-1

[33] A.R. Sarkar, Y.-S. Chuang, N. Mohammed, X. Jiang, De-identification is not enough: a comparison between de-identified and synthetic clinical notes, Sci. Rep. 14 (2024) 29669.

[34] L. Bonomi, S. Gousheh, L. Fan, Enabling health data sharing with fine-grained privacy, in: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, CIKM '23, Association for Computing Machinery, New York, NY, USA, 2023, pp. 131–141, https://doi.org/10.1145/3583780.3614864

[35] R. Catelli, F. Gargiulo, V. Casola, G. De Pietro, H. Fujita, M. Esposito, A novel Covid-19 data set and an effective deep learning approach for the de-identification of Italian medical records, IEEE Access 9 (2021) 19097–19110, https://doi.org/10.1109/ACCESS.2021.3054479

[36] X. Yang, T. Lyu, Q. Li, C.-Y. Lee, J. Bian, W.R. Hogan, Y. Wu, A study of deep learning methods for de-identification of clinical notes in cross-institute settings, in: BMC MEDICAL INFORMATICS AND DECISION MAKING, 2nd International Workshop on Health Natural Language Processing (HealthNLP), vol. 19, Xian, PEOPLES R CHINA, JUN, 10, 2019 2019, https://doi.org/10.1186/s12911-019-0935-4.

[37] C. El-Hayek, S. Barzegar, N. Faux, K. Doyle, P. Pillai, S.J. Mutch, A. Vaisey, R. Ward, L. Sanci, A.G. Dunn, M.E. Hellard, J.S. Hocking, K. Verspoor, D.I.R. Boyle, An evaluation of existing text de-identification tools for use with patient progress notes from Australian general practice, Int. J. Med. Inform. 173 (2023) https://doi.org/10.1016/j.ijmedinf.2023.105021

[38] P.M. Heider, S.M. Meystre, An extensible evaluation framework applied to clinical text deidentification natural language processing tools: multisystem and multicorpus study, J. Med. Internet Res. 26 (2024) https://doi.org/10.2196/55676

[39] M.C. Langenbach, B. Foldyna, I. Hadzic, I.L. Langenbach, V.K. Raghu, M.T. Lu, T.G. Neilan, J.C. Heemelaar, Automated anonymization of radiology reports: comparison of publicly available natural language processing and large language models, Eur. Radiol. (2024) https://doi.org/10.1007/s00330-024-11148-x

[40] L. Liu, O. Perez-Concha, A. Nguyen, V. Bennett, L. Jorm, De-identifying Australian hospital discharge summaries: an end-to-end framework using ensemble of deep learning models, J. Biomed. Inform. 135 (2022) https://doi.org/10.1016/j.jbi.2022.104215

[41] P. Richter-Pechanski, A. Amr, H.A. Katus, C. Dieterich, Deep learning approaches outperform conventional strategies in de-identification of German medical reports, in: R. Rohrig, H. Binder, H. Prokosch, U. Sax, I. Schmidtmann, S. Stolpe, A. Zapf (Eds.), GERMAN MEDICAL DATA SCIENCES: SHAPING CHANGE - CREATIVE SOLUTIONS FOR INNOVATIVE MEDICINE (GMDS 2019), Volume 267 of Studies in Health Technology and Informatics, German Assoc Med Informat Biometry & Epidemiol E V, IOS PRESS, NIEUWE HEMWEG 6B, 1013 BG AMSTERDAM, NETHERLANDS, 08-11 September 2019, 2019, pp. 101–109, https://doi.org/10.3233/SHTI190813, 64th Annual Meeting of the German-Association-of-Medical-Informatics-Biometry-and-Epidemiology (GMDS e V) – Shaping Change – Creative Solutions for Innovative Medicine, Dortmund, GERMANY.

[42] D.P. Kotevski, M. Field, K. Broadley, C.M. Vajdic, R.I. Smee, Y.N. Nemes, Evaluation of an automated presidio anonymisation model for unstructured radiation oncology electronic medical records in an Australian setting, Int. J. Med. Inform. 168 (2022) https://doi.org/10.1016/j.ijmedinf.2022.104880
[43] A. Redd, S. Pickard, S. Meystre, J. Scehnet, D. Bolton, J. Heavirland, A.L. Weaver, C. Hope, J.H. Garvin, Evaluation of PHI Hunter in natural language processing research, Perspect. Health Inf. Manag. 12 (2015) 1f.
[44] A. Henriksson, M. Kvist, H. Dalianis, Prevalence estimation of protected health information in Swedish clinical text, In R. Randell, R. Cornet, C. McCowan, N. Peek, P. Scott (Eds.), INFORMATICS FOR HEALTH: CONNECTED CITIZEN-LED WELLNESS AND POPULATION HEALTH of Studies in Health Technology and Informatics, vol. 235, IOS PRESS, NIEUWE HEMWEG 6B, 1013 BG, AMSTERDAM, NETHERLANDS, 2017, pp. 216–220, https://doi.org/10.3233/978-1-61499-753-5-216
[45] Z. Jian, X. Guo, S. Liu, H. Ma, S. Zhang, R. Zhang, J. Lei, A cascaded approach for Chinese clinical text de-identification with less annotation effort, J. Biomed. Inform. 73 (2017) 76–83, https://doi.org/10.1016/j.jbi.2017.07.017
[46] L. Radhakrishnan, G. Schenk, K. Muenzen, B. Oskotsky, H.A. Choshali, T. Plunkett, S. Israni, A.J. Butte, A certified de-identification system for all clinical text documents for information extraction at scale, JAMIA Open 6 (2023) https://doi.org/10.1093/jamiaopen/ooad045
[47] L. Du, C. Xia, Z. Deng, G. Lu, S. Xia, J. Ma, A machine learning based approach to identify protected health information in Chinese clinical text, Int. J. Med. Inform. 116 (2018) 24–32, https://doi.org/10.1016/j.ijmedinf.2018.05.010
[48] R. Juez-Hernandez, L. Quijano-Sanchez, F. Liberatore, J. Gomez, Agora: an intelligent system for the anonymization, information extraction and automatic mapping of sensitive documents, Appl. Soft Comput. 145 (2023) https://doi.org/10.1016/j.asoc.2023.110540
[49] H.A. Xu, V. Loftsson, B. Kulynych, B. Kaabachi, J.L. Raisaro, Accelerating clinical text annotation in underrepresented languages: a case study on text DE-identification, Stud. Health Technol. Inform. 316 (2024) 853–857.
[50] H. Zhou, D. Ruan, An embedding-based medical note de-identification approach with minimal annotation, in: 2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE), 2020, pp. 263–268, https://doi.org/10.1109/BIBE50027.2020.00050
[51] P.J. Chambon, C. Wu, J.M. Steinkamp, J. Adleberg, T.S. Cook, C.P. Langlotz, Automated deidentification of radiology reports combining transformer and "hide in plain sight" rule-based methods, J. Am. Med. Inform. Assoc. 30 (2023) 318–328, https://doi.org/10.1093/jamia/ocac219
[52] Y. Kim, P.M. Heider, S.M. Meystre, Comparative study of various approaches for ensemble-based DE-identification of electronic health record narratives, AMIA ANNU, Symp. Proc. 2020 (2020) 648–657.
[53] V. Foufi, C. Gaudet-Blavignac, R. Chevrier, C. Lovis, De-identification of medical narrative data, in: R. Engelbrecht, R. Balicer, M. HercigonjaSzekeres (Eds.), PRACTICE OF PATIENT CENTERED CARE: EMPOWERING AND ENGAGING PATIENTS IN THE DIGITAL ERA Of Studies in Health Technology and Informatics, European Federat Med Informat; Israelian Assoc Med Informat, IOS PRESS, NIEUWE HEMWEG 6B, 1013 BG, European Federation for Medical Informatics Special Topic Conference (EFMI STC), vol. 244, AMSTERDAM, NETHERLANDS, 2017, Tel Aviv, ISRAEL,, Oct 2017, pp. 23–27, https://doi.org/10.3233/978-1-61499-824-2-23
[54] K. Kajiyama, H. Horiguchi, T. Okumura, M. Morita, Y. Kano, De-identifying free text of Japanese electronic health records, J. Biomed. Semantics 11 (2020) 11.
[55] X. Tannier, P. Wajsburt, A. Calliger, B. Dura, A. Mouchet, M. Hilka, R. Bey, Development and validation of a natural language processing algorithm to pseudonymize documents in the context of a clinical data warehouse, Methods Inf. Med. 63 (2024) 21–34, https://doi.org/10.1055/s-0044-1778693
[56] Y. Tchouka, J.-F. Couchot, D. Laiymani, P. Selles, A. Rahmani, Differentially private de-identifying textual medical document is compliant with challenging NLP analyses: example of privacy-preserving Icd-10 code association, Intell. Syst. Appl. 23 (2024) https://doi.org/10.1016/j.iswa.2024.200416
[57] H. Horng, J. Steinkamp, C.E. Kahn Jr., T.S. Cook, Ensemble approaches to recognize protected health information in radiology reports, J. Digit. Imaging 35 (2022) 1694–1698, https://doi.org/10.1007/s10278-022-00673-0
[58] T. Paul, M.K.Z. Rana, P.A. Tautam, T.V.P. Kotapati, Y. Jampani, N. Singh, H. Islam, V. Mandhadi, V. Sharma, M. Barnes, R.D. Hammer, A.S.M. Mosa, Investigation of the utility of features in a clinical de-identification model: a demonstration using ehr pathology reports for advanced Nsclc patients, Front. Digit. Health 4 (2022) https://doi.org/10.3389/fdgth.2022.728922
[59] J. Liu, S. Gupta, A. Chen, C.-K. Wang, P. Mishra, H.-J. Dai, Z.S.-Y. Wong, J. Jonnagaddala, Opendeid pipeline for unstructured electronic health record text notes based on rules and transformers: deidentification algorithm development and validation study, J. Med. Internet Res. 25 (2023) https://doi.org/10.2196/48145
[60] C.-K. Wang, F.-D. Wang, Y.-Q. Lee, P.-T. Chen, B.-H. Wang, C.-H. Su, J.-C.-C. Kuo, C.-S. Wu, Y.-L. Chien, H.-J. Dai, V.S. Tseng, W.-L. Hsu, Principle-based approach for the de-identification of code-mixed electronic health records, IEEE Access 10 (2022) 22875–22885, https://doi.org/10.1109/ACCESS.2022.3148396
[61] Y.-Q. Lee, C.-T. Chen, C.-C. Chen, C.-H. Lee, P. Chen, C.-S. Wu, H.-J. Dai, Unlocking the secrets behind advanced artificial intelligence language models in deidentifying chinese-English mixed clinical text: development and validation study, J. Med. Internet Res. 26 (2024), https://doi.org/10.2196/48443
[62] Z. Liu, Y. Chen, B. Tang, X. Wang, Q. Chen, H. Li, J. Wang, Q. Deng, S. Zhu, Automatic de-identification of electronic medical records using token-level and character-level conditional random fields, J. Biomed. Inform. 58 (2015) S47–S52, https://doi.org/10.1016/j.jbi.2015.06.009
[63] I. Perez-Diez, R. Perez-Moraga, A. Lopez-Cerdan, J.-M. Salinas-Serrano, M. de la Iglesia-Vaya, De-identifying spanish medical texts-named entity recognition applied to radiology reports, J. Biomed. Semant. 12 (2021) https://doi.org/10.1186/s13326-021-00236-2
[64] C.A. Libbi, J. Trienes, D. Trieschnigg, C. Seifert, Generating synthetic training data for supervised de-identification of electronic health records, FUTURE INTERNET 13 (2021) https://doi.org/10.3390/fi13050136
[65] Y.-Q. Lee, B.-H. Wang, C.-H. Su, P.-T. Chen, W.-Q. Lin, C.-S. Wu, H.-J. Dai, Protected health information recognition of unstructured code-mixed electronic health records in Taiwan, Stud. Health Technol. Inform. 290 (2022) 627–631.
[66] R. Jackson, I. Kartoglu, C. Stringer, G. Gorrell, A. Roberts, X. Song, H. Wu, A. Agrawal, K. Lui, T. Groza, D. Lewsley, D. Northwood, A. Folarin, R. Stewart, R. Dobson, Cogstack - experiences of deploying integrated information retrieval and extraction services in a large National Health Service Foundation Trust hospital, BMC Med. Inform. Decis. Mak. 18 (2018) https://doi.org/10.1186/s12911-018-0623-9
[67] S. Lima, N. Perez, L. Garcia-Sardina, M. Cuadros, Hitzalmed: anonymisation of clinical text in Spanish, in: N. Calzolari, F. Bechet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, S. Piperidis (Eds.), PROCEEDINGS OF THE 12TH INTERNATIONAL CONFERENCE ON LANGUAGE RESOURCES AND EVALUATION (LREC 2020), EUROPEAN LANGUAGE RESOURCES ASSOC-ELRA, 55-57, RUE BRILLAT-SAVARIN, PARIS, 75013, FRANCE, MAY 11-16, 2020, 2020, Marseille, FRANCE, pp. 7038–7043.
[68] A. Garcia-Pablos, N. Perez, M. Cuadros, Sensitive data detection and classification in spanish clinical text: experiments with BERT, in: N. Calzolari, F. Bechet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, S. Piperidis (Eds.), Proceedings of the 12th International Conference on Language Resources and Evaluation (LREC 2020), EUROPEAN LANGUAGE RESOURCES ASSOC-ELRA, 55-57, RUE BRILLAT-SAVARIN, PARIS, 75013, FRANCE, 2020. 12th International Conference on Language Resources and Evaluation (LREC), Marseille, FRANCE, 11-16 May 2020, pp. 4486–4494.
[69] H. Dalianis, S. Velupillai, De-identifying Swedish clinical text - refinement of a gold standard and experiments with conditional random fields, J. Biomed. Semant. 1 (2010) 6, https://doi.org/10.1186/2041-1480-1-6
[70] H. Berg, H. Dalianis, A semi-supervised approach for de-identification of Swedish clinical text, in: N. Calzolari, F. Bechet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, S. Piperidis (Eds.), PROCEEDINGS OF THE 12TH INTERNATIONAL CONFERENCE ON LANGUAGE RESOURCES AND EVALUATION (LREC 2020), EUROPEAN LANGUAGE RESOURCES ASSOC-ELRA, 55-57, RUE BRILLAT-SAVARIN, PARIS, 75013, FRANCE, 2020. 12th International Conference on Language Resources and Evaluation (LREC), Marseille, FRANCE, 11-16 May 2020, 2020 pp. 4444–4450.
[71] T. Chomutare, K.Y. Yigzaw, A. Budrionis, A. Makhlysheva, F. Godtliebsen, H. Dalianis, De-identifying Swedish ehr text using public resources in the general domain, in: L. PapeHaugaard, C. Lovis, I. Madsen, P. Weber, P. Nielsen, P. Scott (Eds.), DIGITAL PERSONALIZED HEALTH AND MEDICINE Of Studies in Health Technology and Informatics, IOS PRESS, NIEUWE HEMWEG 6B, 1013 BG, 30th Medical Informatics Europe (MIE) Conference, European Federat Med Informat, vol. 270, AMSTERDAM, NETHERLANDS, 2020, Geneva, SWITZERLAND, APR,, 2020, pp. 148–152, https://doi.org/10.3233/SHTI200140
[72] A. Lamproudis, S. Mora, T.O. Svenning, T. Torsvik, T. Chomutare, P.D. Ngo, H. Dalianis, De-identifying Norwegian clinical text using resources from Swedish and danish, in: AMIA Annu. Symp. Proc. 2023, 2023, pp. 456–464.
[73] T. Vakili, A. Lamproudis, A. Henriksson, H. Dalianis, Downstream task performance of BERT models pre-trained using automatically de-identified clinical data, in: N. Calzolari, F. Bechet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, H. Mazo, H. Odijk, S. Piperidis (Eds.), LREC 2022: THIRTEEN INTERNATIONAL CONFERENCE ON LANGUAGE RESOURCES AND EVALUATION, Google; S African CTR Digital Language Resources; Vocapia Res; 3M; Emvista; Expert.ai; Grammarly; Minist Culture, Delegat Gen Langue Francaise Aux Langues France; Reg Sud Provence Alpes Cote Azur, EUROPEAN LANGUAGE RESOURCES ASSOC-ELRA, 55-57, RUE BRILLAT-SAVARIN, PARIS, 75013, FRANCE, 2022, Marseille, FRANCE, 20-25 Jun 2022, pp. 4245–4252. 13th International Conference on Language Resources and Evaluation (LREC).
[74] S.-Y. Shin, Y.R. Park, Y. Shin, H.J. Choi, J. Park, Y. Lyu, M.-S. Lee, C.-M. Choi, W.-S. Kim, J.H. Lee, A de-identification method for bilingual clinical texts of various note types, J. Korean Med. Sci. 30 (2015) 7–15, https://doi.org/10.3346/jkms.2015.30.1.7
[75] P. Richter-Pechanski, S. Riezler, C. Dieterich, De-identification of German medical admission notes, Stud. Health Technol. Inform. 253 (2018) 165–169.
[76] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (1997) 1735–1780, https://doi.org/10.1162/neco.1997.9.8.1735
[77] J.D. Lafferty, A. McCallum, F.C.N. Pereira, Conditional random fields: probabilistic models for segmenting and labeling sequence data, in: Proceedings of the Eighteenth International Conference on Machine Learning, ICML '01, San Francisco, CA, USA, Morgan Kaufmann Publishers Inc, 2001, pp. 282–289.
[78] J.A. OpenAI, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F.L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, R. Avila, I. Babuschkin, S. Balaji, V. Balcom, P. Baltescu, H. Bao, M. Bavarian, J. Belgum, I. Bello, J. Berdine, G. Bernadett-Shapiro, C. Berner, L. Bogdonoff, O. Boiko, M. Boyd, A.-L. Brakman, G. Brockman, T. Brooks, M. Brundage, K. Button, T. Cai, R. Campbell, A. Cann, B. Carey, C. Carlson, R. Carmichael, B. Chan, C. Chang, F. Chantzis, D. Chen, S. Chen, R. Chen, J. Chen, M. Chen, B. Chess, C. Cho, C. Chu, H.W. Chung, D. Cummings, J. Currier, Y. Dai, C. Decareaux, T. Degry, N. Deutsch, D. Deville, A. Dhar, D. Dohan, S. Dowling,

S. Dunning, A. Ecoffet, A. Eleti, T. Eloundou, D. Farhi, L. Fedus, N. Felix, S.P. Fishman, J. Forte, I. Fulford, L. Gao, E. Georges, C. Gibson, V. Goel, T. Gogineni, G. Goh, R. Gontijo-Lopes, J. Gordon, M. Grafstein, S. Gray, R. Greene, J. Gross, S.S. Gu, Y. Guo, C. Hallacy, J. Han, J. Harris, Y. He, M. Heaton, J. Heidecke, C. Hesse, A. Hickey, W. Hickey, P. Hoeschele, B. Houghton, K. Hsu, S. Hu, X. Hu, J. Huizinga, S. Jain, S. Jain, J. Jang, A. Jiang, R. Jiang, H. Jin, D. Jin, S. Jomoto, B. Jonn, H. Jun, T. Kaftan, A.K. Łukasz Kaiser, I. Kanitscheider, N.S. Keskar, T. Khan, L. Kilpatrick, J.W. Kim, C. Kim, Y. Kim, J.H. Kirchner, J. Kiros, M. Knight, D. Kokotajlo, A.K. Łukasz Kondraciuk, A. Konstantinidis, K. Kosic, G. Krueger, V. Kuo, M. Lampe, I. Lan, T. Lee, J. Leike, J. Leung, D. Levy, C.M. Li, R. Lim, M. Lin, S. Lin, M. Litwin, T. Lopez, R. Lowe, P. Lue, A. Makanju, K. Malfacini, S. Manning, T. Markov, Y. Markovski, B. Martin, K. Mayer, A. Mayne, B. McGrew, S.M. McKinney, C. McLeavey, P. McMillan, J. McNeil, D. Medina, A. Mehta, J. Menick, L. Metz, A. Mishchenko, P. Mishkin, V. Monaco, E. Morikawa, D. Mossing, T. Mu, M. Murati, O. Murk, D. Mély, A. Nair, R. Nakano, R. Nayak, A. Neelakantan, R. Ngo, H. Noh, L. Ouyang, C. O'Keefe, J. Pachocki, A. Paino, J. Palermo, A. Pantuliano, G. Parascandolo, J. Parish, E. Parparita, A. Passos, M. Pavlov, A. Peng, A. Perelman, F. de Avila Belbute Peres, M. Petrov, H.P. de Oliveira Pinto, P. Michael, M. Pokrass, V.H. Pong, T. Powell, A. Power, B. Power, E. Proehl, R. Puri, A. Radford, J. Rae, A. Ramesh, C. Raymond, F. Real, K. Rimbach, C. Ross, B. Rotsted, H. Roussez, N. Ryder, M. Saltarelli, T. Sanders, S. Santurkar, G. Sastry, H. Schmidt, D. Schnurr, J. Schulman, D. Selsam, K. Sheppard, T. Sherbakov, J. Shieh, S. Shoker, P. Shyam, S. Sidor, E. Sigler, M. Simens, J. Sitkin, K. Slama, I. Sohl, B. Sokolowsky, Y. Song, N. Staudacher, F.P. Such, N. Summers, I. Sutskever, J. Tang, N. Tezak, M.B. Thompson, P. Tillet, A. Tootoonchian, E. Tseng, P. Tuggle, N. Turley, J. Tworek, J.F.C. Uribe, A. Vallone, A. Vijayvergiya, C. Voss, C. Wainwright, J.J. Wang, A. Wang, B. Wang, J. Ward, J. Wei, C. Weinmann, A. Welihinda, P. Welinder, J. Weng, L. Weng, M. Wiethoff, D. Willner, C. Winter, S. Wolrich, H. Wong, L. Workman, S. Wu, J. Wu, M. Wu, K. Xiao, T. Xu, S. Yoo, K. Yu, Q. Yuan, W. Zaremba, R. Zellers, C. Zhang, M. Zhang, S. Zhao, T. Zheng, J. Zhuang, W. Zhuk, B. Zoph, Gpt-4 technical report, arXiv:2303.08774, 2024.

[79] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, G. Lample, Llama: Open and efficient foundation language models, arXiv:2302.13971, 2023.

[80] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: J. Burstein, C. Doran, T. Solorio (Eds.), Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), Minneapolis, Minnesota, Association for Computational Linguistics, 2019, pp. 4171–4186, https://doi.org/10.18653/v1/N19-1423

[81] W. Rahman, M.K. Hasan, A. Zadeh, L. Morency, M.E. Hoque, M-BERT: injecting multimodal information in the BERT structure, CoRR arXiv:1908.05787, 2019.

[82] V. Menger, F. Scheepers, L.M. van Wijk, M. Spruit, Deduce: a pattern matching method for automatic de-identification of Dutch medical text, Telemat. Inform. 35 (2018) 727–736, https://doi.org/10.1016/j.tele.2017.08.002

[83] K. Kajiyama, H. Horiguchi, T. Okumura, M. Morita, Y. Kano, De-identifying free text of Japanese electronic health records, J. Biomed. Semant. 11 (2020) https://doi.org/10.1186/s13326-020-00227-9

[84] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. U. Kaiser, I. Polosukhin, in: I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett (Eds.), Attention is all you need, Advances in Neural Information Processing Systems, volume 30, Curran Associates, Inc., 2017, https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

[85] D. Bahdanau, K. Cho, Y. Bengio, Neural machine translation by jointly learning to align and translate, in: 3rd International Conference on Learning Representations, ICLR 2015; Conference Date: 07-05-2015 Through 09-05-2015, 2015.

[86] P. Wang, Y. Li, L. Yang, S. Li, L. Li, Z. Zhao, S. Long, F. Wang, H. Wang, Y. Li, C. Wang, An efficient method for deidentifying protected health information in Chinese electronic health records: algorithm development and validation, J. M. I. R. Med. Inform. 10 (2022) https://doi.org/10.2196/38154

[87] R. Catelli, F. Gargiulo, V. Casola, G. De Pietro, H. Fujita, M. Esposito, A novel Covid-19 data set and an effective deep learning approach for the de-identification of Italian medical records, IEEE Access 9 (2021) 19097–19110, https://doi.org/10.1109/ACCESS.2021.3054479

[88] Y.S. Yashwanth, R. Shettar, Zero and few short learning using large language models for de-identification of medical records, IEEE Access 12 (2024) 110385–110393, https://doi.org/10.1109/ACCESS.2024.3439680

[89] B. Singh, Q. Sun, Y.S. Koh, J. Lee, E. Zhang, Detecting protected health information with an incremental learning ensemble: a case study on New Zealand clinical text, in: G. Webb, Z. Zhang, V. Tseng, G. Williams, M. Vlachos, L. Cao (Eds.), 2020 IEEE 7TH INTERNATIONAL CONFERENCE ON DATA SCIENCE AND ADVANCED ANALYTICS (DSAA 2020), Proceedings of the International Conference on Data Science and Advanced Analytics, IEEE; IEEE Comp Soc; IEEE Computat Intelligence Soc; Amer Stat Assoc; Assoc Comp Machinery; ACM SIGKDD; China Comp Confederat; Macquarie Univ; Monash Univ; Business Events Sydney, IEEE COMPUTER SOC, 10662 LOS VAQUEROS CIRCLE, PO BOX 3014, LOS ALAMITOS, CA 90720-1264 USA, 7th IEEE International Conference on Data Science and Advanced Analytics (DSAA), Univ Technol Sydney, ELECTR NETWORK, OCT 06-09, 2020 2020, pp. 719–728, https://doi.org/10.1109/DSAA49011.2020.00082.

[90] B. Tang, D. Jiang, Q. Chen, X. Wang, J. Yan, Y. Shen, De-identification of clinical text via bi-LSTM-CRF with neural language models, AMIA Annu. Symp. Proc. 2019 (2019) 857–863.

[91] R. Catelli, V. Casola, G. De Pietro, H. Fujita, M. Esposito, Combining contextualized word representation and sub-document level analysis through bi-lstm plus CRF architecture for clinical de-identification, Knowl.-Based Syst. 213 (2021) https://doi.org/10.1016/j.knosys.2020.106649

[92] T. Hartman, M.D. Howell, J. Dean, S. Hoory, R. Slyper, I. Laish, O. Gilon, D. Vainstein, G. Corrado, K. Chou, M.J. Po, J. Williams, S. Ellis, G. Bee, A. Hassidim, R. Amira, R.G. Beryozkin, I. Szpektor, Y. Matias, Customization scenarios for de-identification of clinical notes, BMC Med. Inform. Decis. Mak. 20 (2020) https://doi.org/10.1186/s12911-020-1026-2

[93] K. Murugadoss, A. Rajasekharan, B. Malin, V. Agarwal, S. Bade, J.R. Anderson, J.L. Ross, W.A. Faubion, Jr., J.D. Halamka, V. Soundararajan, S. Ardhanari, Building a best-in-class automated de-identification tool for electronic health records through ensemble learning, PATTERNS 2 (2021) https://doi.org/10.1016/j.patter.2021.100255

[94] B. Norgeot, K. Muenzen, T.A. Peterson, X. Fan, B.S. Glicksberg, G. Schenk, E. Rutenberg, B. Oskotsky, M. Sirota, J. Yazdany, G. Schmajuk, D. Ludwig, T. Goldstein, A.J. Butte, Protected health information filter (philter): accurately and securely de-identifying free-text clinical notes, NPJ Digit. Med. 3 (2020) 57, https://doi.org/10.1038/s41746-020-0258-y

[95] M. Kayaalp, A.C. Browne, Z.A. Dodd, P. Sagan, C.J. McDonald, De-identification of address, date, and alphanumeric identifiers in narrative clinical reports, AMIA Annu. Symp. Proc. 2014 (2014) 767–776.

[96] J.M. Steinkamp, T. Pomeranz, J. Adleberg, C.E. Kahn, Jr., T.S. Cook, Evaluation of automated public de-identification tools on a corpus of radiology reports, RADIOL.-ARTIF. INTELL. 2 (2020) https://doi.org/10.1148/ryai.2020190137

[97] I. Neamatullah, M.M. Douglass, L.-W.H. Lehman, A. Reisner, M. Villarroel, W.J. Long, P. Szolovits, G.B. Moody, R.G. Mark, G.D. Clifford, Automated de-identification of free-text medical records, BMC Med. Inform. Decis. Mak. 8 (2008) 32, https://doi.org/10.1186/1472-6947-8-32

[98] Y. Kim, P. Heider, S. Meystre, Ensemble-based methods to improve DE-identification of electronic health record narratives, AMIA Annu. Symp. Proc. 2018 (2018) 663–672.

[99] J. Aberdeen, S. Bayer, R. Yeniterzi, B. Wellner, C. Clark, D. Hanauer, B. Malin, L. Hirschman, The Mitre identification scrubber toolkit: design, training, and assessment, Int. J. Med. Inform. 79 (2010) 849–859, https://doi.org/10.1016/j.ijmedinf.2010.09.007, https://www.sciencedirect.com/science/article/pii/S1386505610001681.

[100] F. Dernoncourt, J.Y. Lee, P. Szolovits, Neuroner: an easy-to-use program for named-entity recognition based on neural networks, in: L. Specia, M. Post, M. Paul (Eds.), Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Association for Computational Linguistics, Copenhagen, Denmark, 2017, pp. 97–102, https://doi.org/10.18653/v1/D17-2017

[101] J. Gardner, L. Xiong, An integrated framework for de-identifying unstructured medical data, data & knowledge engineering 68, in: Including Special Section: 21st IEEE International Symposium on Computer-Based Medical Systems (IEEE CBMS 2008) – Seven Selected and Extended Papers on Biomedical Data Mining, 2009, pp. 1441–1451, https://doi.org/10.1016/j.datak.2009.07.006, https://www.sciencedirect.com/science/article/pii/S0169023X09001074.

[102] B. Ribeiro, V. Rolla, R. Santos, Incognitus: a toolbox for automated clinical notes anonymization, in: D. Croce, L. Soldaini (Eds.), 17TH CONFERENCE OF THE EUROPEAN CHAPTER OF THE ASSOCIATION FOR COMPUTATIONAL LINGUISTICS, EACL 2023, Assoc Computat Linguist, European Chapter; Grammarly; Liveperson; Amazon Sci; Bloomberg; Duolingo; Adobe; Babelscape, ASSOC COMPUTATIONAL LINGUISTICS-ACL, 209 N EIGHTH STREET, STROUDSBURG, PA 18360 USA, May 02-04 2023, 2023, pp. 187–194. 17th Conference of the European-Chapter of the Association-for-Computational-Linguistics (EACL) – System Demonstrations, Dubrovnik, CROATIA.

[103] P. Vcelak, M. Kryl, M. Kratochvíl, J. Klecková, Identification and classification of DICOM files with burned-in text content, Int. J. Med. Inform. 126 (2019) 128–137, https://doi.org/10.1016/J.IJMEDINF.2019.02.011

[104] E. Monteiro, C. Costa, J.L. Oliveira, A de-identification pipeline for ultrasound medical images in dicom format, J. Med. Syst. 41 (2017) https://doi.org/10.1007/s10916-017-0736-1

[105] T.E.D. Campos, B.R. Babu, M. Varma, M. Varma, Character recognition in natural images, in: Proceedings of the International Conference on Computer Vision Theory and Applications, Lisbon, Portugal, 2009, https://www.microsoft.com/en-us/research/publication/character-recognition-in-natural-images/.

[106] R. Smith, An overview of the tesseract OCR engine, in: ICDAR '07: Proceedings of the Ninth International Conference on Document Analysis and Recognition, Washington, DC, USA, IEEE Computer Society, 2007, pp. 629–633, https://storage.googleapis.com/pub-tools-public-publication-data/pdf/33418.pdf.

[107] J.L. Leevy, T.M. Khoshgoftaar, F. Villanustre, Survey on RNN and CRF models for de-identification of medical free text, J. Big Data 7 (2020) 73, https://doi.org/10.1186/s40537-020-00351-4

[108] L. Hafsa, S. Kozeta, Preserving privacy in medical images while still enabling ai-driven research: a comprehensive review, in: 2024 13th Mediterranean Conference on Embedded Computing (MECO), 2024, pp. 1–5, https://doi.org/10.1109/MECO62516.2024.10577795

[109] K. Lee, M. Kayaalp, S. Henry, O. Uzuner, A context-enhanced de-identification system, ACM Trans. Comput. Healthcare 3 (2021) https://doi.org/10.1145/3470980

[110] S.-Y. Shin, Y.R. Park, Y. Shin, H.J. Choi, J. Park, Y. Lyu, M.-S. Lee, C.-M. Choi, W.-S. Kim, J.H. Lee, A de-identification method for bilingual clinical texts of various note types, J. Korean Med. Sci. 30 (2015) 7–15.

[111] H.-J. Lee, Y. Wu, Y. Zhang, J. Xu, H. Xu, K. Roberts, A hybrid approach to automatic de-identification of psychiatric notes, J. Biomed. Inform. 75 (2017) S19–S27, https://doi.org/10.1016/j.jbi.2017.06.006

[112] M. Friedrich, A. Köhn, G. Wiedemann, C. Biemann, Adversarial learning of privacy-preserving text representations for de-identification of medical records, in: A. Korhonen, D. Traum, L. Màrquez (Eds.), Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics, Florence, Italy, Association for Computational Linguistics, 2019, pp. 5829–5839, https://doi.org/10.18653/v1/P19-1584

[113] K. Lee, M. Filannino, O. Uzuner, An empirical test of grus and deep contextualized word representations on de-identification, in: L. OhnoMachado, B. Seroussi (Eds.), MEDINFO 2019: HEALTH AND WELLBEING E-NETWORKS FOR ALL, volume 264 Studies in Health Technology and Informatics, French Assoc Med Informat, 17th World Congress of Medical and Health Informatics (MEDINFO), Int Med Informat Assoc, Lyon, FRANCE, IOS PRESS, NIEUWE HEMWEG 6B, 1013 BG, 25-30 Aug 2019, pp. 218–222, https://doi.org/10.3233/SHTI190215 AMSTERDAM, NETHERLANDS, 2019.

[114] P. Wang, Y. Li, L. Yang, S. Li, L. Li, Z. Zhao, S. Long, F. Wang, H. Wang, Y. Li, C. Wang, An efficient method for deidentifying protected health information in Chinese electronic health records: algorithm development and validation, JMIR Med. Inform. 10 (2022) e38154.

[115] M.E. Azzouzi, G. Coatrieux, R. Bellafqira, D. Delamarre, C. Riou, N. Oubenali, S. Cabon, M. Cuggia, G. Bouzille, Automatic de-identification of French electronic health records: a cost-effective approach exploiting distant supervision and deep learning models, BMC Med. Inform. Decis. Mak. 24 (2024) https://doi.org/10.1186/s12911-024-02422-5

[116] H. Yang, J.M. Garibaldi, Automatic detection of protected health information from clinic narratives, J. Biomed. Inform. (2015) S30–S38, https://doi.org/10.1016/j.jbi.2015.06.015

[117] B. He, Y. Guan, J. Cheng, K. Cen, W. Hua, CRFS based de-identification of medical records, J. Biomed. Inform. 58 (2015) S39–S46, https://doi.org/10.1016/j.jbi.2015.08.012

[118] A. Dehghan, A. Kovacevic, G. Karystianis, J.A. Keane, G. Nenadic, Combining knowledge- and data-driven methods for de-identification of clinical narratives, J. Biomed. Inform. 58 (2015) S53–S59, https://doi.org/10.1016/j.jbi.2015.06.029

[119] Z. Liu, B. Tang, X. Wang, Q. Chen, De-identification of clinical notes via recurrent neural network and conditional random field, J. Biomed. Inform. 75 (2017) S34–S42, https://doi.org/10.1016/j.jbi.2017.05.023

[120] T. Chomutare, K.Y. Yigzaw, A. Budrionis, A. Makhlysheva, F. Godtliebsen, H. Dalianis, De-identifying Swedish EHR text using public resources in the general domain, Stud. Health Technol. Inform. 270 (2020) 148–152.

[121] P. Burckhardt, R. Padman, Deidentify, AMIA Annu. Symp. Proc. 2017 (2017) 485–494.

[122] B. Singh, Q. Sun, Y.S. Koh, J. Lee, E. Zhang, Detecting protected health information with an incremental learning ensemble: a case study on New Zealand clinical text, in: 2020 IEEE 7th International Conference on Data Science and Advanced Analytics (DSAA), 2020, pp. 719–728, https://doi.org/10.1109/DSAA49011.2020.00082

[123] Z. Liu, M. Yang, X. Wang, Q. Chen, B. Tang, Z. Wang, H. Xu, Entity recognition from clinical texts via recurrent neural network, BMC Med. Inform. Decis. Mak. 17 (2017) https://doi.org/10.1186/s12911-017-0468-7

[124] J.M. Steinkamp, T. Pomeranz, J. Adleberg, C.E. Kahn, Jr, T.S. Cook, Evaluation of automated public DE-identification tools on a corpus of radiology reports, Radiol. Artif. Intell. 2 (2020) e190137.

[125] F. Chen, S.M.A. Bokhari, K. Cato, G. Gursoy, S. Rossetti, Examining the generalizability of pretrained de-identification transformer models on narrative nursing notes, Appl. Clin. Inform. 15 (2024) 357–367, https://doi.org/10.1055/a-2282-4340

[126] J. Trienes, D. Trieschnigg, C. Seifert, D. Hiemstra, Comparing rule-based, feature-based and deep neural methods for de-identification of Dutch medical records, in: Proc. Health Search and Data Mining Workshop, HSDM, 2020.

[127] R.G. Andrzejak, K. Lehnertz, F. Mormann, C. Rieke, P. David, C.E. Elger, Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity: dependence on recording region and brain state, Phys. Rev. E 64 (2001) 061907, https://doi.org/10.1103/PhysRevE.64.061907

[128] R. Quinlan, Thyroid disease, UCI Mach. Learn. Repos. (1986) https://doi.org/10.24432/C5D010

[129] B. Strack, J.P. DeShazo, C. Gennings, J.L. Olmo, S. Ventura, K.J. Cios, J.N. Clore, Impact of hba1c measurement on hospital readmission rates: analysis of 70,000 clinical database patient records, BioMed Res. Int. 2014 (2014) 781670, https://doi.org/10.1155/2014/781670

[130] A. Dehghan, A. Kovacevic, G. Karystianis, J.A. Keane, G. Nenadic, Learning to identify protected health information by integrating knowledge- and data-driven algorithms: a case study on psychiatric evaluation notes, J. Biomed. Inform. 75 (2017) S28–S33, https://doi.org/10.1016/j.jbi.2017.06.005

[131] H.-J. Lee, Y. Zhang, K. Roberts, H. Xu, Leveraging existing corpora for de-identification of psychiatric notes using domain adaptation, AMIA ANNU, Symp. Proc. 2017 (2017) 1070–1079.

[132] Y.-S. Zhao, K.-L. Zhang, H.-C. Ma, K. Li, Leveraging text skeleton for de-identification of electronic medical records, BMC MEDICAL INFORMATICS AND DECISION MAKING 18 (2018), in: 3rd China Health Information Processing Conference (CHIP), Shenzhen, PEOPLES R CHINA, 24-25 Nov 2017, https://doi.org/10.1186/s12911-018-0598-6

[133] C. Iwendi, S.A. Moqurrab, A. Anjum, S. Khan, S. Mohan, G. Srivastava, N-sanitization: a semantic privacy-preserving framework for unstructured medical datasets, Comput. Commun. 161 (2020) 160–171, https://doi.org/10.1016/j.comcom.2020.07.032

[134] S.H. Lee, Natural language generation for electronic health records, NPJ Digit. Med. 1 (2018) https://doi.org/10.1038/s41746-018-0070-0

[135] D.D.A. Bui, M. Wyatt, J.J. Cimino, The UAB informatics Institute and 2016 CEGS n-GRID de-identification shared task challenge, J. Biomed. Inform. 75S (2017) S54–S61.

[136] M. Abdalla, M. Abdalla, F. Rudzicz, G. Hirst, Using word embeddings to improve the privacy of clinical notes, J. Am. Med. Inform. Assoc. 27 (2020) 901–907, https://doi.org/10.1093/jamia/ocaa038

[137] N. Le, Leveraging transformers-based language models in proteome Bioinformatics, Proteomics. 23 (6) (2023) e2300011.

[138] T. Tran, N. Le, SA-TTCA: an SVM-based approach for tumor t-cell antigen classification using features extracted from biological sequencing and natural language processing, Comput. Biol. Med. 174 (4) (2024) 108408.